

FIG 1 Research path as visual argument

Supported claim: component-token separation beats fixed physics-prior escalation for aggregate VPP forecasting.

a — Mechanism: replace the shared cancellation channel

The figure separates the rejected shared-encoder route from the supported A1/iGT route.

REJECTED FAMILY



$$P_{net} = L - PV - W + B$$

signed errors can
cancel in aggregate

SUPPORTED FAMILY



A1: 8 variable tokens, no fixed physics priors, net-MSE supervision only.

b — Primary outcome

Test MAE, $\times 10^{-3}$ MW. Lower is better.

12.5%

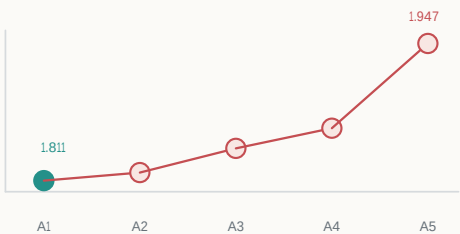
lower mean MAE



~20x lower seed s.d. across three seeds.

c — Negative control chain

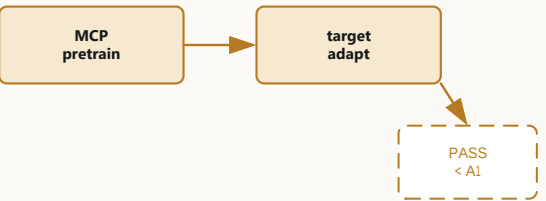
Fixed priors improve Val but worsen Test.



C12: complexity is an overfitting amplifier here.

d — Open gate, not a result

Phase B remains deliberately amber.



Gate threshold: A1 mean Test MAE = 1.811×10^{-3} .

Do not cite Phase B as evidence until repaired protocol and target-portfolio tests pass.

Claim binding: C08-C12 are supported; Phase B is shown as a gated research branch, not a reported result.

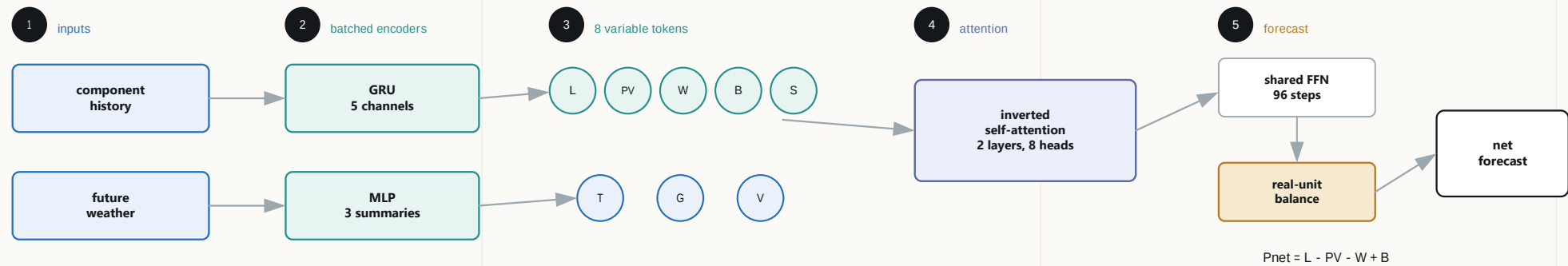
FIG 2

Editable A1 architecture and training branch

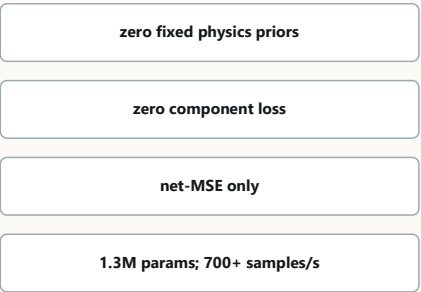
The main model stays intentionally simple; rejected additions and Phase-B training are subordinate annotations.

a — A1/iGT main path

Five component histories and three weather summaries become variable tokens before inverted attention.



b — Model contract



This contract is the key thesis constraint, not a placeholder.

c — Rejected additions

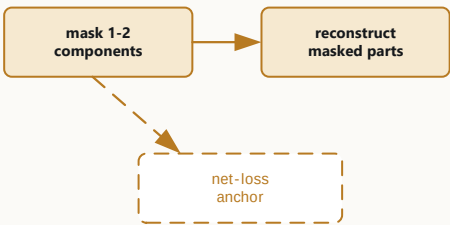


+ physics token -> + twin/constraint tokens -> + graph bias -> + horizon decoder

Test MAE worsens monotonically: 1.819, 1.843, 1.863, 1.947.

Interpretation: stronger Val gains predict worse Test transfer.

d — Phase-B training branch



Shown as a branch because target-portfolio adaptation remains unresolved.

Abbreviations: L, load; PV, photovoltaic; W, wind; B, battery power; S, battery SOC; T, temperature; G, irradiance; V, wind speed.